

Notes: Ichniowski, Shaw, and Prennushi (AER, 1997)

The Effects of Human Resource Management Practices on Productivity: A Study of Steel Finishing Lines

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
2.1	Contribution relative to prior HRM and productivity studies	3
2.2	Contribution relative to incentive and organizational theory	4
2.3	Contribution relative to selection-bias concerns	4
2.4	What the paper adds to management and labor economics	4
3	Data and measurement	4
3.1	Sample design	4
3.2	Production technology and productivity outcome	5
3.3	Technology and input controls	5
3.4	HRM practice measures	6
3.5	HRM systems	6
4	Models and empirical specifications	6
4.1	Baseline productivity model	6
4.2	Fixed-effects model	7
4.3	Robustness specifications	7
4.4	Quality model	7
4.5	Individual practice regressions	7
5	Identification and empirical design	8
5.1	Cross-sectional identification	8
5.2	Within-line identification	8
5.3	Why the design is credible	8
5.4	What the paper cannot fully prove	8
6	Tables	9
6.1	Table 1: HRM variable definitions	9
6.2	Table 2 and Table 3: HRM systems and raw productivity differences	9
6.3	Table 4: Main productivity estimates	10
6.4	Table 5: Fixed-effects robustness checks	11
6.5	Table 6: Quality outcome	11
6.6	Table 7: HRM systems versus individual practices	11
6.7	Appendix Table A1: full control-variable estimates	12
7	Short summary	13
8	Conclusion	14
8.1	Core conclusion	14

8.2	Economic intuition	14
8.3	Takeaway	14

1 Big picture

- **Question.** Do innovative human resource management practices raise productivity, and do they work mainly as isolated practices or as complementary systems?
- **Setting.** The paper studies steel finishing lines, a narrowly defined manufacturing process in which output and line technology are highly comparable across establishments.
- **Data advantage.** The authors personally visit production sites, interview managers, workers, and union representatives, and collect monthly production data, detailed engineering controls, and HR practice data.
- **Sample.** The main panel contains up to 2,190 monthly observations on 36 steel finishing lines owned by 17 companies.
- **Outcome.** The main productivity measure is *uptime*, the fraction of scheduled production time during which the line actually runs. Since potential output is mechanically determined by width, gauge, speed, and scheduled hours, uptime is a clean productivity measure for this process.
- **Main result.** Innovative HRM systems are associated with substantially higher productivity. The strongest systems combine incentives, careful selection, teamwork, job rotation, employment security, training, and communication.
- **Complementarity result.** Systems of practices matter more than individual practices. Adding single HRM practices to the regression generally does not explain the productivity differences once HRM systems are included.
- **Fixed-effects evidence.** Lines that switch from traditional to more innovative HRM systems experience productivity gains, so the result is not solely driven by cross-sectional differences across lines.
- **Robustness.** Results survive detailed engineering controls, controls for management tenure, manager-specific dummies, shutdown-threat variables, year dummies, and quality-output specifications.
- **Economic takeaway.** Workplace organization is a production input. The productivity effect of HRM depends on coherent bundles of mutually reinforcing practices, not simply on adopting isolated “best practices.”

2 Incremental contribution of the paper

2.1 Contribution relative to prior HRM and productivity studies

Prior empirical studies often examined one practice at a time, such as profit sharing, training, information sharing, or grievance procedures. Many used establishment-level, firm-level, or cross-industry data. This paper contributes by using a very narrowly defined production process and by measuring many HR practices jointly.

- It reduces production heterogeneity by focusing on one type of steel finishing line rather than on broad industries or heterogeneous plants.
- It provides unusually detailed controls for technology and input quality based on engineering interviews and site visits.
- It measures entire HRM systems rather than only individual practices.
- It combines cross-sectional variation with within-line changes in HRM systems over time.

Interpretation: The paper improves credibility on two margins at once: better measurement of the production function and better measurement of the organizational practices.

2.2 Contribution relative to incentive and organizational theory

Theoretical work by Holmstrom and Milgrom, Milgrom and Roberts, and related organizational economics models stresses complementarity among practices: incentives, job design, training, information sharing, and decision rights may reinforce one another. The paper gives empirical support for this idea.

- The best-performing lines do not merely have one incentive plan or one training practice; they have a coherent high-performance work system.
- Individual practices are often weak once systems are included, consistent with the view that practices are valuable when combined.
- The ordering of HRM systems is monotonic: more comprehensive systems have higher productivity.

Interpretation: The incremental contribution is not just that HR practices matter; it is that *complementary bundles* of HR practices matter.

2.3 Contribution relative to selection-bias concerns

A standard worry is that better firms choose better HR systems, so cross-sectional associations overstate the causal effect of HRM. The paper addresses this concern by exploiting within-line changes.

- Fixed-effects regressions compare a line to itself before and after HRM changes.
- Control variables absorb time-varying technology, input quality, start-up periods, and equipment changes.
- Additional models include controls for prechange productivity growth, manager tenure, manager identities, threat effects, and year dummies.

Interpretation: The design is not a randomized experiment, but it is stronger than a simple cross-section because the same production line can be observed under different HRM systems.

2.4 What the paper adds to management and labor economics

The paper connects labor economics, industrial relations, and production economics.

- It shows that workplace practices have economically meaningful productivity effects in a capital-intensive manufacturing setting.
- It demonstrates that unionized and nonunionized lines can both be represented in a common productivity model.
- It links HRM practices not only to labor outcomes but directly to physical production outcomes.
- It provides an early empirical foundation for the high-performance work systems literature.

Interpretation: The paper is important because it translates qualitative claims about modern manufacturing and worker involvement into measurable productivity estimates.

3 Data and measurement

3.1 Sample design

- The authors focus on one specific steel finishing process to reduce heterogeneity in output and technology.
- They personally visited 45 lines owned by 21 companies.
- Of these, 41 lines had performance data, and 36 provided comparable monthly productivity data.

- The final productivity sample includes 36 lines owned by 17 companies and up to 2,190 line-month observations.
- The sample includes union and nonunion lines, large and small steel producers, high and low productivity lines, and different HRM environments.

Interpretation: The narrow process focus is central to the paper. By comparing similar production lines, the authors avoid many problems that arise when comparing firms across different technologies or products.

3.2 Production technology and productivity outcome

A steel finishing line processes a continuous strip of flat-rolled steel. The coil is loaded, unwound, treated, recoiled, and cut. Potential output is determined by technical parameters:

$$\text{Potential } Q_{it} = f(w_{it}, g_{it}, s_{it}, h_{it}^s), \quad (1)$$

where w is width, g is gauge, s is speed, and h^s is scheduled hours.

Actual output is:

$$\text{Actual } Q_{it} = f(w_{it}, g_{it}, s_{it}, h_{it}^s) \times (1 - d_{it}), \quad (2)$$

where d_{it} is the fraction of scheduled hours lost to unscheduled line stops.

The paper therefore defines productivity as:

$$Uptime_{it} = 1 - d_{it}. \quad (3)$$

Interpretation: Uptime is a particularly clean productivity measure because, conditional on engineering parameters and product mix, production depends on how often the line actually runs rather than on hard-to-measure output quality differences.

3.3 Technology and input controls

The productivity equations include many controls for non-HRM determinants of uptime:

- line age and line age squared;
- year built and year built squared;
- start-up period indicators and start-up time trends;
- steel input quality;
- scheduled maintenance shifts;
- customer type;
- maximum line speed and width;
- specific equipment dummies;
- computerization of line operations;
- installation periods for new equipment.

Interpretation: These controls are unusually detailed for an HRM productivity study. They are included to separate organizational effects from engineering differences.

3.4 HRM practice measures

The paper measures practices in seven broad HRM policy areas:

1. incentive pay;
2. recruiting and selection;
3. teamwork;
4. employment security;
5. flexible job assignment;
6. skills training;
7. communication.

It also records traditional labor relations variables, such as union status and grievance rates.

Interpretation: The HRM data are not generic survey items. They are based on interviews, documents, and site-specific evidence about how the production line is actually operated.

3.5 HRM systems

The authors group lines into four HRM systems.

- **System 1:** the most innovative system. It combines line incentives, careful screening, high participation, multiple teams, formal teams, employment security, job rotation, training, and communication.
- **System 2:** an intermediate innovative system with extensive teams, training, and communication, but missing some high-performance practices.
- **System 3:** a partial innovation system emphasizing formal teams and communication, with fewer incentives and less training.
- **System 4:** the traditional system with narrow jobs, close supervision, strict work rules, hourly pay, little worker involvement, and little formal training.

Interpretation: These systems are ordered from most to least comprehensive. The main hypothesis is that productivity improves as lines move toward more coherent innovative systems.

4 Models and empirical specifications

4.1 Baseline productivity model

The baseline uptime equation is:

$$U_{it} = \gamma' H_{it} + \beta' X_{it} + \varepsilon_{it}, \quad (4)$$

where:

- U_{it} is percent uptime for line i in month t ;
- H_{it} is a vector of HRM system indicators;
- X_{it} is a vector of technology, input, and production controls;
- γ measures productivity differences associated with HRM systems.

Interpretation: The baseline model compares productivity across lines with different HRM systems after controlling for observed engineering differences.

4.2 Fixed-effects model

To address time-invariant line-specific heterogeneity, the authors estimate:

$$U_{it} - \bar{U}_i = \gamma'(H_{it} - \bar{H}_i) + \beta'(X_{it} - \bar{X}_i) + (\varepsilon_{it} - \bar{\varepsilon}_i). \quad (5)$$

- Fixed effects remove permanent productivity differences across lines.
- Identification comes from lines that change HRM systems.
- The model estimates how productivity changes when a line moves to a new HRM system.

Interpretation: This is the closest design to a within-line before-after comparison with controls. It helps address the concern that inherently better lines selected into better HRM systems.

4.3 Robustness specifications

The paper augments the fixed-effects model with additional time-varying controls:

- management tenure variables;
- manager-specific dummy variables;
- shutdown threat variables;
- year dummies;
- prechange productivity growth controls.

Interpretation: These specifications test whether HRM changes proxy for changes in management, closure threats, time trends, or regression-to-the-mean dynamics.

4.4 Quality model

The authors also estimate models for prime yield, a quality measure:

$$PrimeYield_{it} = \gamma'H_{it} + \beta'X_{it} + \varepsilon_{it}. \quad (6)$$

Interpretation: If innovative HRM systems merely raise output by sacrificing quality, the quality regression would show weak or negative effects. Instead, HRM systems are associated with higher quality as well.

4.5 Individual practice regressions

To test complementarity, the paper estimates models that include both HRM system indicators and individual HRM practice variables.

$$U_{it} = \gamma'H_{it} + \delta'P_{it} + \beta'X_{it} + \varepsilon_{it}, \quad (7)$$

where P_{it} is the vector of individual HRM practices.

Interpretation: If single practices matter independently, individual practice coefficients should remain large and robust. If complementarities matter, system indicators should dominate and individual practice coefficients should be weak or unstable.

5 Identification and empirical design

5.1 Cross-sectional identification

The OLS estimates use differences across production lines in HRM systems, while controlling for observed technological differences.

- Advantage: uses a large number of line-month observations.
- Limitation: lines may select HRM systems based on unobserved productivity potential.

Interpretation: OLS provides the first evidence that more innovative HRM systems have higher productivity, but selection bias remains possible.

5.2 Within-line identification

The fixed-effects estimates use changes in HRM systems within the same line over time.

- Advantage: controls for all permanent line-specific differences.
- Limitation: HRM adoption may coincide with other time-varying changes.

Interpretation: The paper strengthens identification by adding controls for time-varying technology, managers, shutdown threats, and productivity trends.

5.3 Why the design is credible

The design is credible because:

- production processes are highly comparable;
- productivity is precisely measured;
- the authors observe detailed technology controls;
- HRM practices are collected from site visits and documents;
- fixed effects exploit within-line HRM changes.

Interpretation: No single element fully solves selection, but together they make the productivity comparisons much more convincing than typical cross-firm HRM studies.

5.4 What the paper cannot fully prove

- HRM systems are not randomly assigned.
- The sample is one industry and one production process.
- The paper cannot fully separate all dimensions of HRM bundles.
- The number of lines that change systems is limited.
- Results may not mechanically generalize to services or less standardized manufacturing.

Interpretation: The paper provides strong evidence for this setting and a powerful test of complementarity, but its external validity must be evaluated carefully.

6 Tables

6.1 Table 1: HRM variable definitions

Read:

- Table 1 defines the individual HRM variables used to characterize each production line.
- The most common practices are profit sharing, information sharing, and worker meetings.
- Less common practices include job rotation, high screening, high training, and line incentives.
- Union status is very common in the sample, with a mean of 0.917.

Economic message: Table 1 shows that HRM systems are constructed from observable practices across incentive pay, selection, teams, security, flexibility, training, and communication.

TABLE 1—DEFINITIONS OF HRM VARIABLES

HRM variable name	Mean	Dummy variable description
<i>1. Incentive pay</i>		
a. Profit sharing	0.700	Is there a company profit-sharing plan covering the line workers?
b. Line incentives	0.186	Are operators covered by a "nontraditional" incentive pay plan which applies across shifts of workers and which is sensitive to quality as well as quantity aspects of output?
<i>2. Recruiting and selection</i>		
a. High screening	0.085	Was an extensive selection procedure used to hire new workers, including tests for personality traits needed for cooperative team environments and efforts to set clear expectations about required work behaviors of the new workers?
<i>3. Teamwork</i>		
a. High participation	0.237	Are a majority of operators involved in formal or informal work teams or other related problem-solving activities?
b. Multiple teams	0.130	Do operators participate in more than one problem-solving team?
c. Formal team practice	0.335	Are operators organized into formal work teams either on the line or for the purposes of problem-solving activities according to an established policy with at least some operators involved in team activities?
<i>4. Employment security</i>		
a. Employment security	0.288	Has the company committed to a goal of long-term employment security and offered employees a pledge of employment security?
<i>5. Flexible job assignment</i>		
a. Job rotation	0.079	Do operators rotate across jobs or tasks on the line?
<i>6. Skills training</i>		
a. High train	0.134	Have all operators on the line received off-the-job training?
b. Low train	0.208	Have at least some operators received off-the-job training?
<i>7. Communication</i>		
a. Information sharing	0.566	Are operators and union representatives, if any, provided with financial information on a regular basis?
b. Meet workers	0.508	Do line managers meet off-line with operators to discuss issues of concern, including issues related to performance and quality?
c. Meet union	0.224	Do union representatives and managers meet often to discuss concerns and cooperate in finding solutions to issues?
<i>8. Labor relations</i>		
a. Union	0.917	Is the line a unionized operation?
b. Low grievance	0.499	Is the grievance filing rate less than 12 per year?

Notes: Means reported in column 2 refer to the means for the main sample of $N = 2,190$ line-months used in the productivity models reported in Tables 4 and 5. For the meet union and low grievance variables, we assign nonunion lines a value of 1 for these two dummy variables because of regular meetings with worker representatives and low levels of complaints in their formal or informal complaint or grievance procedures. The means of these two variables among the sample of union observations are 0.153 and 0.453, respectively.

6.2 Table 2 and Table 3: HRM systems and raw productivity differences

Read:

- Table 2 shows the practice composition of the four HRM systems.
- System 1 includes nearly all innovative practices; System 4 includes none of the measured innovative practices.

- System 2 is close to System 1 but lacks some practices such as extensive screening or employment security.
- System 3 is a partial system built mainly around teams and communication.
- Table 3 shows raw average uptime by starting and ending HRM system.
- Lines in more innovative systems have higher average uptime, and switchers often show improvements after upgrading HRM systems.

Economic message: The descriptive evidence already suggests a hierarchy: more comprehensive HRM systems are associated with higher productivity.

TABLE 2—PROPORTION OF PRODUCTION LINES WITH SPECIFIC HRM PRACTICES
WITHIN FOUR HRM SYSTEM CATEGORIES^a

Practices in seven HRM policy areas ^b	HRM System 1	HRM System 2	HRM System 3	HRM System 4
1. <i>Incentive pay</i>				
a. Line incentives	1.00	0.31	0.00	0.00
2. <i>Recruiting and selection</i>				
a. High screening	1.00	0.15	0.00	0.00
3. <i>Teamwork</i>				
a. High participation	1.00	0.85	0.10	0.00
b. Multiple teams	1.00	0.62	0.00	0.00
c. Formal team practice	1.00	1.00	1.00	0.00
4. <i>Employment security</i>				
a. Employment security	1.00	0.23	0.48	0.00
5. <i>Flexible job assignment</i>				
a. Job rotation	1.00	0.15	0.03	0.00
6. <i>Skills training</i>				
a. High train	1.00	0.69	0.00	0.00
b. Low train	1.00	0.92	0.07	0.00
7. <i>Communication</i>				
a. Information sharing	1.00	0.54	0.62	0.00
b. Meet workers	1.00	0.77	0.72	0.00

^a The sample for Table 2 is based on production lines and not production-line months. This sample includes 54 observations, and not just 36 production lines, because, for lines that switch HRM practices, it includes one observation for each different combination of HRM practices that the lines experience.

^b See Table 1 for definitions of HRM variables.

Table 2: HRM system categories

TABLE 3—THE DISTRIBUTION OF SAMPLE OBSERVATIONS AND AVERAGE PRODUCTIVITIES BY HRM SYSTEM

Starting HRM system	Ending HRM system			
	HRM System 4	HRM System 3	HRM System 2	HRM System 1
HRM System 4	<i>Uptime in System 4</i> 0.899 (0.036) [N = 782]	<i>Uptime in System 3</i> 0.912 (0.039)	<i>Uptime in System 2</i> 0.939 (0.021)	—
		<i>Prior uptime (Sys. 4)</i> 0.901 (0.034) [N = 172]	<i>Prior uptime (Sys. 3)</i> 0.912 (0.028)	
			<i>Prior uptime (Sys. 4)</i> 0.894 (0.081) [N = 59]	
HRM System 3	—	<i>Uptime in System 3</i> 0.930 (0.032) [N = 742]	<i>Uptime in System 2</i> 0.964 (0.011)	—
			<i>Prior uptime (Sys. 3)</i> 0.949 (0.027) [N = 82]	
HRM System 2	—	—	<i>Uptime in System 2</i> 0.924 (0.070) [N = 287]	—
HRM System 1	—	—	—	<i>Uptime in System 1</i> 0.940 (0.041) [N = 77]

Note: Standard deviations are in parentheses (); sample size for cell is in brackets [].

Table 3: observations and productivity by system

6.3 Table 4: Main productivity estimates

Read:

- OLS estimates with detailed technology controls show that System 1 lines have uptime about 6.7 to 7.8 percentage points higher than traditional System 4 lines.
- System 2 coefficients are about 3.2 to 4.1 percentage points.
- System 3 coefficients are about 1.4 to 2.5 percentage points.
- Fixed-effects models show positive productivity gains when lines adopt more innovative systems.
- Relative to System 4, adopting System 2 raises uptime by about 3.5 percentage points in column (3), and adopting System 3 raises uptime by about 2.5 percentage points.
- With controls for prechange productivity growth, the System 2 and System 3 coefficients rise to 6.8 and 4.3 percentage points.

Economic message: The main table supports both the level effect and the within-line improvement effect of innovative HRM systems.

6.4 Table 5: Fixed-effects robustness checks

Read:

- Table 5 adds controls for management tenure, manager-specific effects, shutdown threats, and year dummies.
- System 2 and System 3 coefficients remain positive and statistically significant across all specifications.
- The most complete model has coefficients of 5.2 percentage points for System 2 and 4.4 percentage points for System 3.
- The robustness checks reduce the concern that HRM effects simply proxy for better managers or crisis-driven effort.

Economic message: The fixed-effects results are not fragile. The HRM system effects survive a demanding set of time-varying controls.

TABLE 4—ESTIMATED PRODUCTIVITY EFFECTS OF HRM SYSTEMS IN OLS AND FIXED-EFFECTS MODELS
(DEPENDENT VARIABLE: PERCENT UPTIME)
[N = 2,190]

HRM system	OLS models without detailed technology controls ^a		OLS models with detailed technology controls ^b		Fixed effects models ^c	
	Classification of HRM systems from Table 2 (1a)	Classification of HRM systems from Noninate procedure (1b)	Classification of HRM systems from Table 2 (2a)	Classification of HRM systems from Noninate procedure (2b)	No controls for prechange productivity growth (3)	With controls for prechange productivity growth (4)
1. HRM System 1	0.097*** (0.007)	0.114*** (0.006)	0.067*** (0.007)	0.078*** (0.007)	—	—
2. HRM System 2	0.038*** (0.004)	0.057*** (0.004)	0.032*** (0.005)	0.041*** (0.005)	0.035*** (0.008)	0.068*** (0.019)
3. HRM System 3	0.011*** (0.002)	0.029*** (0.003)	0.014*** (0.003)	0.025*** (0.006)	0.025*** (0.006)	0.043*** (0.011)
R ²	0.246	0.283	0.409	0.409	0.066	0.068

^a Control variables in columns (1a)–(1b) are: number of years line has been operating and years squared; year line was built and year built squared; dummy for start-up periods indicated by first 12 months of operations and 1-to-12 time trend for month of start-up operation; 1-to-5 index of quality of steel input; and number of annual eight-hour scheduled maintenance shifts.

^b Control variables in columns (2a)–(2b) are: all controls listed in footnote a; dummy for type of customer; maximum speed of the line and speed squared; maximum width of the line and width squared; nine dummies to indicate specific pieces of equipment from start to finish of the line and a measure of the age of one piece of equipment at end of the line; a dummy to indicate high and low levels of computer control of line operations; and a variable to measure the value of major new equipment during its six-month installation period. For full set of coefficient estimates for the column (2b) model, see Appendix Table A1.

^c Results for fixed-effects models are identical under different HRM classification procedures because all procedures identify the same lines as lines that switch HRM systems. There are no coefficient estimates for HRM System 1 in the fixed-effects model since no lines switched into this system. Other control variables in columns (3) and (4) are: age of line and age squared; dummy for start-up periods indicated by first 12 months of operations and 1-to-12 time trend for month of start-up operation; 1-to-5 index of quality of steel input; age of the end-of-the-line piece of equipment; and a variable to measure the value of major new equipment during its six-month installation period. For full set of coefficients for the column (3) model, see Appendix Table A1.

*** Significant at the 0.01 level.

TABLE 5—ESTIMATED PRODUCTIVITY EFFECTS OF HRM SYSTEMS IN FIXED-EFFECTS MODELS WITH CONTROLS FOR TIME-VARYING DETERMINANTS OF PRODUCTIVITY
(DEPENDENT VARIABLE: PERCENT UPTIME)
[N = 2,190]

HRM system	(1)	(2)	(3)	(4)	(5)	(6)
1. HRM System 2	0.035*** (0.008)	0.046*** (0.009)	0.064*** (0.012)	0.034*** (0.009)	0.034*** (0.009)	0.052*** (0.014)
2. HRM System 3	0.025*** (0.006)	0.026*** (0.006)	0.048*** (0.010)	0.025*** (0.006)	0.025*** (0.006)	0.044*** (0.011)
<i>Time-varying controls</i>						
3a. Technology controls ^a	Yes	Yes	Yes	Yes	Yes	Yes
3b. Management tenure variables ^b	No	Yes	No	No	No	Yes
3c. Manager-specific dummy variables ^c	No	No	Yes	No	No	Yes
3d. Shutdown threat variables ^d	No	No	No	Yes	No	Yes
3e. Year dummy variables	No	No	No	No	Yes	Yes
F tests ^e	F(6,2175) = 9.074 F(72,2109) = 3.95 F(2,2174) = 7.27 F(13,2171) = 5.93 F(93,2088) = 4.03					
R ²	0.066	0.090	0.176	0.072	0.099	0.199

^a The technology variables are listed in footnote c to Table 4 [the Table 5, column (1), model replicates the Table 4, column (3), model].

^b The management tenure variables are the tenures of the HR and line managers (and squared terms) and dummy variables for tenure less than three years for those managers.

^c The management-specific dummy variables are 72 dummy variables equal to one for each specific HR manager and line manager team in place.

^d The threat variables are the percent of the plant permanently shut down and a dummy for the recent use of layoffs.

^e The F-tests test the joint null hypothesis that the variables added in that column have coefficients that are significantly different from

6.5 Table 6: Quality outcome

Read:

- Table 6 uses percent prime yield as the dependent variable.
- System 1 has a large positive quality effect in OLS: 15.2 percentage points without detailed controls and 13.2 with detailed controls.
- System 2 and System 3 also have positive effects on quality.
- Fixed-effects estimates show positive and significant quality improvements for System 2 and System 3.

Economic message: The productivity gains are not achieved by sacrificing quality. More innovative HRM systems improve both uptime and prime yield.

6.6 Table 7: HRM systems versus individual practices

Read:

- Each row adds one individual HRM practice to the model along with the HRM system indicators.
- HRM system coefficients remain positive and significant when individual practices are added.
- Individual practice coefficients are often small, unstable, or insignificant in fixed-effects models.

- Some practices look positive in OLS but lose independent explanatory power once systems and fixed effects are included.

Economic message: Table 7 is the paper's key complementarity evidence. The productivity effect belongs to systems of practices, not simply to isolated practices.

TABLE 6—ESTIMATED EFFECTS OF HRM SYSTEMS ON QUALITY OF PRODUCTION IN OLS AND FIXED-EFFECTS MODELS
(DEPENDENT VARIABLE: PERCENT PRIME YIELD)
[N = 1,750]

	OLS models without detailed machinery controls ^a	OLS models with detailed machinery controls ^b	Fixed-effects models ^c
HRM measure	(1)	(2)	(3)
1a. HRM System 1	0.152*** (0.008)	0.132*** (0.009)	—
1b. HRM System 2	0.098*** (0.005)	0.046*** (0.007)	0.036*** (0.007)
1c. HRM System 3	0.064*** (0.004)	0.044*** (0.005)	0.050*** (0.005)

^a Control variables in column (1) are: number of years line has been operating and years squared; year line was built and year built squared; dummy for start-up periods indicated by first 12 months of operations and 1-to-12 time trend for month of start-up operation; 1-to-5 index of quality of steel input; number of annual eight-hour scheduled maintenance shifts; and four dummies for slight differences in how prime yield is measured in different lines.

^b Control variables in column (2) are: all controls listed in footnote a; dummy for type of customer; maximum speed of the line and speed squared; maximum width of the line and width squared; nine dummies to indicate specific pieces of equipment from start to finish of the line and a measure of the age of one piece of equipment at end of the line; a dummy to indicate high and low levels of computer control of line operations; a variable to measure the value of major new equipment during its six-month installation period; and four dummies for slight differences in how prime yield is measured in different lines.

^c Control variables in column (3) are: age of line and age squared; dummy for start-up periods indicated by first 12 months of operations and 1-to-12 time trend for month of start-up operation; 1-to-5 index of quality of steel input; age of the end-of-the-line piece of equipment; and a variable to measure the value of major new equipment during its six-month installation period.

*** Significant at the 0.01 level.

TABLE 7—ESTIMATED PRODUCTIVITY EFFECTS OF HRM SYSTEMS AND INDIVIDUAL HRM PRACTICES
(DEPENDENT VARIABLE: PERCENT UPTIME) [N = 2,190]^a

Individual HRM practice included in models in the row	OLS models with HRM system variables and with individual HRM practice variable ^b				Fixed-effects models with HRM system variables and with individual HRM practice variable ^{c,d}			OLS models with only individual HRM practice ^e	Fixed-effects models with only individual HRM practice ^e
	Coef. on HRM System 1	Coef. on HRM System 2	Coef. on HRM System 3	Coef. on individual HRM practice	Coef. on HRM System 2	Coef. on HRM System 3	Coef. on individual HRM practice	Coef. on individual HRM practice	Coef. on individual HRM practice
	(1a)	(1b)	(1c)	(1d)	(2a)	(2b)	(2c)	(3)	(4)
1. Profit sharing	0.066*** (0.007)	0.026*** (0.005)	0.012*** (0.003)	0.006** (0.003)	—	—	—	0.007** (0.002)	—
2. Line incentives	0.068*** (0.007)	0.032*** (0.005)	0.014*** (0.003)	−0.002 (0.004)	0.031*** (0.009)	0.023*** (0.006)	0.007 (0.005)	0.012*** (0.004)	0.012*** (0.004)
3. High screening	0.068*** (0.015)	0.032*** (0.005)	0.013*** (0.003)	−0.001 (0.001)	—	—	—	0.029*** (0.004)	—
4. High participation	0.095*** (0.009)	0.052*** (0.006)	0.019*** (0.004)	−0.023** (0.004)	0.058*** (0.010)	0.028*** (0.006)	−0.024*** (0.006)	0.005* (0.003)	−0.008* (0.005)
5. Formal team practice	0.062*** (0.008)	0.027*** (0.006)	0.012*** (0.004)	0.004 (0.003)	0.034*** (0.013)	0.025*** (0.008)	0.001 (0.006)	0.014*** (0.003)	0.011*** (0.004)
6. Multiple teams	0.060*** (0.011)	0.026*** (0.008)	0.014*** (0.009)	0.008 (0.009)	—	—	—	0.026*** (0.004)	0.010* (0.006)
7. Employment security	0.047*** (0.010)	0.023*** (0.005)	0.008** (0.004)	0.011 (0.004)	—	—	—	0.025*** (0.003)	—
8. Job rotation	0.084*** (0.009)	0.033*** (0.005)	0.018*** (0.004)	−0.021*** (0.006)	—	—	—	0.010** (0.005)	—
9. High train	0.068*** (0.008)	0.033*** (0.006)	0.014*** (0.003)	−0.003 (0.006)	0.037*** (0.009)	0.025*** (0.006)	−0.006 (0.007)	0.016*** (0.004)	0.001 (0.007)
10. Low train	0.054*** (0.009)	0.021*** (0.007)	0.013*** (0.003)	0.013** (0.006)	0.028*** (0.010)	0.023*** (0.006)	0.008 (0.006)	0.026*** (0.004)	0.015*** (0.005)
11. Information sharing	0.077*** (0.011)	0.038*** (0.007)	0.017*** (0.004)	−0.004 (0.003)	—	—	—	0.013*** (0.002)	0.010* (0.006)
12. Meet workers	0.060*** (0.008)	0.028*** (0.005)	0.012*** (0.004)	0.005 (0.004)	0.028*** (0.010)	0.024*** (0.006)	0.007 (0.006)	0.021*** (0.003)	0.012*** (0.004)
13. Meet union	0.067*** (0.010)	0.032*** (0.006)	0.014*** (0.004)	0.00003 (0.004)	0.031*** (0.009)	0.025*** (0.006)	0.014 (0.014)	0.017*** (0.003)	0.024** (0.012)
14. Union	0.073*** (0.008)	0.037*** (0.006)	0.016*** (0.004)	0.012 (0.008)	—	—	—	−0.026*** (0.007)	—
15. Low grievance	0.066*** (0.008)	0.031*** (0.005)	0.014*** (0.004)	0.0003 (0.003)	0.013 (0.013)	0.008 (0.010)	0.022** (0.010)	0.012*** (0.003)	0.029*** (0.006)

Notes: To interpret this table, note that each row represents up to four regressions. For example, in row 2 the coefficients in columns (1a)–(1d) are from one OLS regression containing three system variables and one individual practice variable (for line incentives); the coefficients in columns (2a)–(2c) are from the comparable fixed-effects regression; the coefficient in column (3) is from the OLS regression containing the one practice variable (line incentives) and no system variables; and the coefficient in column (4) is the fixed-effects estimate comparable to column (3).

^a Standard errors reported in parentheses.

^b Other control variables included in the OLS models are those listed in footnote b to Table 4.

^c Other control variables in the fixed-effects models are those listed in footnote c to Table 4.

^d No line switches into HRM System 1 and so the coefficient on HRM System 1 is not estimated in fixed-effects models.

^e * Significant at the 0.10 level ** Significant at the 0.05 level *** Significant at the 0.01 level

6.7 Appendix Table A1: full control-variable estimates

Read:

- Appendix Table A1 reports the full coefficient set behind the detailed OLS and fixed-effects models.
- The table includes HRM system variables, steel input quality, maintenance, line age, start-up effects, equipment dummies, computerization, speed, width, and new equipment installation.
- The HRM system effects remain visible even after these detailed controls are included.

Economic message: Appendix Table A1 confirms that the main results are not artifacts of omitted engineering variables.

APPENDIX

TABLE A1—FULL SET OF COEFFICIENT ESTIMATES OLS AND FIXED-EFFECTS UPTIME MODELS
[FOR COLUMNS (2B) AND (3) OF TABLE 4]

Variable	OLS model with detailed controls	Fixed-effects model
1a. HRM System 1	0.078*** (0.007)	—
1b. HRM System 2	0.041*** (0.005)	0.035*** (0.008)
1c. HRM System 3	0.025*** (0.003)	0.025*** (0.006)
2. Maintenance shifts per year	0.0002*** (0.00004)	—
3. Index of steel input quality	0.008*** (0.002)	0.005** (0.002)
4a. Year built	0.018*** (0.003)	—
4b. (Year built) ²	-0.0001*** (0.00002)	—
5a. Line age	0.0102*** (0.0011)	0.011*** (0.001)
5b. (Line age) ²	-0.0002*** (0.00002)	-0.0002*** (0.00002)
6a. Dummy for start-up period	-0.043** (0.011)	-0.071*** (0.017)
6b. Time trend for start-up period	0.004** (0.001)	0.004** (0.002)
7a. Equipment dummy 1a	0.032*** (0.004)	—
7b. Equipment dummy 1b	0.019*** (0.003)	—
8a. Equipment dummy 2a	-0.043*** (0.005)	—
8b. Equipment dummy 2b	-0.017*** (0.005)	—
8c. Equipment dummy 2c	-0.029*** (0.005)	—
9. Equipment dummy 3	-0.003 (0.002)	—
10. Equipment dummy 4	-0.005 (0.003)	—
11. Equipment dummy 5	0.022** (0.003)	—
12. Age of dummy 5 equipment	-0.0013*** (0.0002)	-0.0018*** (0.0007)
13. Equipment dummy 6	-0.004 (0.006)	—
14. Dummy for low computerization	-0.015*** (0.003)	—
15a. Maximum line speed	-0.00067*** (0.00005)	—
15b. (Maximum line speed) ²	8.1×10^{-7} *** (7×10^{-8})	—

TABLE A1—Continued.

Variable	OLS model with detailed controls	Fixed-effects model
16a. Maximum width	-0.0096*** (0.0011)	—
16b. (Maximum width) ²	0.00010*** (0.00001)	—
17. Dummy for type of end user	-0.039*** (0.006)	—
18. New equipment value during six-month installation	-3.1×10^{-6} *** (9.9×10^{-7})	-2.9×10^{-6} *** (9.6×10^{-7})
19. Intercept	0.376*** (0.075)	—
R ²	0.409	0.073

* Significant at the 0.10 level.
 ** Significant at the 0.05 level.
 *** Significant at the 0.01 level.

Appendix Table A1, part 1

Appendix Table A1, continued

7 Short summary

- The paper studies how HRM systems affect productivity in steel finishing lines.
- It uses a homogeneous production process and detailed site-level data.
- Productivity is measured by uptime, the fraction of scheduled time the line runs.
- HRM practices are measured through interviews, documents, and site visits.
- The authors classify lines into four HRM systems from comprehensive innovative systems to traditional systems.
- OLS estimates show a monotonic productivity hierarchy across HRM systems.
- Fixed-effects models show that lines improve after adopting more innovative HRM systems.
- Results survive detailed controls for technology, managers, shutdown threats, year effects, and prechange productivity growth.
- Quality results show that innovative HRM systems improve prime yield as well as uptime.
- Individual HRM practices do not explain the system effects, supporting complementarity among practices.
- The main contribution is to provide detailed, process-level evidence that high-performance work systems raise productivity.

8 Conclusion

8.1 Core conclusion

Ichniowski, Shaw, and Prennushi show that innovative HRM systems are associated with large productivity gains in steel finishing lines. The strongest systems combine incentive pay, careful selection, teamwork, job flexibility, employment security, training, and communication. These systems raise productivity relative to traditional systems with narrow jobs, strict rules, close supervision, and little worker involvement.

8.2 Economic intuition

The intuition is that modern HRM practices are complementary. Incentives are more powerful when workers have information, skills, and flexibility. Teamwork is more effective when workers are trained and secure enough to share knowledge. Communication is more valuable when employees have discretion to use it. Because the practices reinforce each other, partial adoption produces smaller gains than coherent system adoption.

8.3 Takeaway

The paper is a classic empirical demonstration that workplace organization can be a source of productivity. It also offers one of the clearest early pieces of evidence that organizational complementarities are empirically important. For later work on management practices, high-performance work systems, and firm productivity, the paper provides both a methodological template and a benchmark result.